Boundary constraints for equal-area rectangle dissections

Private recovery manuscript for L018

12 October 2026

A perfect partition in this paper is an axis-parallel dissection of a rectangle into at least two pairwise noncongruent rectangles of equal positive area. Congruence allows a quarter-turn. This paper supplies complete versions of the boundary and search arguments used by García-Colín, Leemans, Müßig and Roldán [1]; the finite exclusions themselves require separately completed computations. No universal nonexistence assertion follows here.

Theorem 1 (Source Lemma 2.1)**.** *A perfect partition has at least seven pieces, even if their side lengths are real.*

Proof. Suppose otherwise and choose such a partition with the least number r of pieces. There cannot be two pieces: they would have equal widths or equal heights and, by equality of area, would be congruent. No piece spans an entire width or height of the container. Removing such a piece leaves one or two nonempty rectangular regions. A region consisting of one piece would make that piece congruent to the removed spanning piece. Every remaining region therefore contains at least two pieces and is a smaller perfect partition, contradicting minimality. Each container side therefore meets at least two pieces along intervals of positive length, and the four corner pieces are distinct.

Suppose each side meets exactly two pieces. These must be its corner pieces, so only the four corner pieces meet the boundary. Scale the two coordinate axes independently to make the container the unit square. This scaling is used only to derive equality of the widths and heights of two pieces, which remains an equality on undoing the scaling. The common normalized area is $a = 1/r$. Let x and y be the widths of the bottom-left and top-left pieces. The widths on the right are $1 - x$ and $1 - y$. Their heights and the two vertical boundary equations give

$$y = \frac{ax}{x - a}, \qquad 1 - y = \frac{a(1 - x)}{1 - x - a}.$$

The denominators are positive because the two lower pieces have heights strictly less than one. Adding the equations and clearing denominators gives $(1-2a)x(1-x) = a(1-2a)$. Since $r > 2$, this implies $x(1 - x) = a$ and $y = 1 - x$. The bottom-left and top-right pieces thus have the same width x and the same height a/x, contrary to noncongruence.

Some side consequently meets $k \geq 3$ pieces. Take this as the bottom side. The heights of these pieces are pairwise distinct: equal heights and equal areas would force congruence. None reaches the top. Each has at least one piece immediately above its upper edge, and these sets of upper neighbors are mutually disjoint, since an upper neighbor's bottom coordinate is the height of the corresponding bottom piece. They also contain no bottom piece. The lowest bottom piece has at least two upper neighbors. Otherwise its unique upper neighbor covers its full top edge. At each endpoint, either the container wall or the adjacent, strictly taller bottom piece prevents horizontal extension, so this neighbor has exactly the same width. Equal area again gives congruence. There are therefore at least $k + (k + 1) \geq 7$ pieces. □

Lemma 2 (Integral positions)**.** *In an axis-parallel tiling with integer piece widths and heights, every piece corner has integer coordinates relative to a container corner.*

Proof. Choose a horizontal line through the interior of a particular vertical piece edge, avoiding all horizontal piece edges. The segments into which the tiling partitions this line have the integer widths of the pieces it crosses. Every endpoint's horizontal coordinate is their partial sum from the left wall. A vertical line gives the analogous conclusion for horizontal edges. □

Proposition 3 (Exact finite search interface)**.** *Fix positive integers W, H, r with $r \mid WH$, and set $A = WH/r$. Let the available shapes be the unordered pairs $\{a, b\}$ with $ab = A$ and at least one orientation fitting the container. A perfect partition into r pieces exists if and only if the following finite skyline search succeeds. Initially all W unit columns have height zero. At each step choose the leftmost column of minimum height y, with horizontal coordinate x. Try every unused shape in each fitting orientation $w \times h$ for which the full interval $[x, x + w)$ has current height y and $y + h \leq H$. Raise that interval by h and continue. Success means r pieces have been placed; otherwise exhaust the choices.*

Proof. Every placement is inside the box and disjoint from earlier placements, and no unordered shape is reused. After r placements the total area is $rA = WH$, so there is no uncovered open region. Conversely, take an actual partition and follow the prescribed order. By Lemma 2, all its coordinates are integer. The piece covering the leftmost lowest uncovered unit cell must begin at (x, y): its bottom cannot lie below the minimum uncovered level, and extending to the left at that level would intersect the covered region or leave the box. The columns across its width must all have the same height y, since a higher one would intersect its interior and a lower one contradicts minimality. Its exact orientation is one of the search choices. This inductively recovers the actual partition. The search has finite depth r and finite branching. □

Lemma 4 (Choice of uncovered valley)**.** *The exact search may instead choose any maximal flat skyline interval whose neighbors have strictly larger covered heights, with container walls allowed as neighbors. In particular it may choose such an interval with fewest currently fitting shapes and orientations.*

Proof. Let the interval be $[x, x + w)$ at height y. In a completion, the piece covering its leftmost lowest uncovered unit cell cannot begin below y, since that would overlap the covered region. It cannot extend to the left because of the wall or the strictly higher neighboring column. It therefore begins at (x, y). It cannot cross the strictly higher right neighbor either. Thus its entire base lies in this flat interval and is one of the legal choices. Each placement still preserves the skyline invariant. Induction on the remaining pieces proves completeness for any deterministic selection of such an interval. A lowest interval always exists until completion. □

Proposition 5 (Safe necessary prunings)**.** *The following conditions may be required at every partial skyline state without losing any completion:*

1. *each remaining open component separated by full-height columns has area a multiple of A;*
2. *the width of every lowest uncovered plateau is the sum of widths of a subset of distinct unused shapes in fitting orientations;*
3. *the four container sides admit side subsets with their exact lengths; adjacent subsets share exactly their common corner piece, and opposite subsets can share only pieces spanning the intervening container dimension.*

For a square, one may additionally require that its bottom-left corner piece has least unordered shape index among all four corner pieces, and that this piece has width at most height. Repeated corner identities must be allowed.

Proof. An unplaced rectangle cannot cross a full-height column, so the first assertion follows by summing equal areas within a component. The pieces starting along a lowest plateau partition

its base interval and have distinct unused shapes, proving the second assertion. The third follows by reading the boundary of an actual tiling; a rectangle meeting two opposite sides spans the intervening width or height. Finally the dihedral symmetries of a square can send a least corner piece to the bottom left. Reflection in the diagonal through that corner then chooses its width at most its height. A piece spanning a dimension may occupy two corners, which is why the least-index inequalities are non-strict. □

Proposition 6 (Parity endpoint constraint)**.** *In any integer tiling, take the parity of the number of uncovered unit cells on each horizontal row of a partial placement. Every remaining rectangle of odd width contributes the indicator of its vertical interval, and rectangles of even width contribute zero. Thus the discrete derivative of the uncovered parity profile is the symmetric difference of the endpoint pairs of the remaining odd-width intervals. The analogous assertion holds for columns and odd heights. Also, the alternating column sum of the uncovered cells is the sum of $\pm h$ over remaining odd-width pieces, and the alternating row sum is the sum of $\pm w$ over remaining odd-height pieces.*

Proof. Reduce the width contributions on a row modulo two and then take differences between consecutive rows, including the two exterior zero rows. An interval's indicator changes exactly at its two endpoints. For the alternating sum, the sum of $(-1)^x$ along an integer interval is zero for even length and is 1 or -1 for odd length. Sum over the height of each rectangle. Columns are dual. □

Proposition 7 (A bounded exact reduction for the hard square)**.** *In a perfect partition of the* 840*-square into* 21 *pieces, a quarter-turn permits the four odd-sided pieces to have widths* $75, 105, 175, 525$ *and respective heights* $64(7, 5, 3, 1)$*. Their bottom coordinates have the form* $t + 64k_i$*, where* $0 \le t$*, and the normalized tuple* (k_1, k_2, k_3, k_4) *is one of*

$$(0,0,4,4), \quad (0,0,5,7), \quad (0,2,0,2),$$
$$(0,3,0,7), \quad (1,0,5,0), \quad (1,3,0,0).$$

This necessary reduction is not an assertion that any such partition exists.

Proof. The smaller factors of 33600 whose partners fit the 840-square are

$$40, 42, 48, 50, 56, 60, 64, 70, 75, 80, 84,$$
$$96, 100, 105, 112, 120, 140, 150, 160, 168, 175.$$

Thus there are exactly 21 shapes of area $840^2/21$, and all must be used. The four with an odd side are

$$(75, 448), \quad (105, 320), \quad (175, 192), \quad (525, 64).$$

All others have two even sides. By the alternating-sum part of Proposition 6, the even partners of those oriented with odd width must have a signed sum zero, and the same holds for those oriented with odd height. After division by 64, the partners are $7, 5, 3, 1$. No nonempty proper subset has a signed zero sum: pairs have unequal members, odd-cardinality subsets have odd sum, and singletons are nonzero. Consequently the four odd dimensions lie along a common axis. Rotate if needed.

The entire square has an even number of cells on every row. Form a multigraph whose vertices are the endpoints of the four odd-width pieces' vertical intervals, and whose edges are those intervals. Every vertex has even degree by Proposition 6. There are no loops. A two-edge component would have equal edge lengths, which the four distinct heights exclude. Hence this four-edge graph is connected and is a four-cycle; any repeated vertex in a closed traversal would split it into two forbidden two-edge cycles. All edge lengths are multiples of 64, so its vertices share one residue modulo 64. The unique partition of $\{7, 5, 3, 1\}$ into equal sums is $7+1 = 5+3$.

Start a traversal with the positively directed length-7 edge; the remaining signed lengths are $1, -5, -3$ in one of their six orders. The six resulting interval-start tuples, after subtracting their minimum, are precisely those displayed. The alternating-column equation additionally forces the starting x parities of pieces 1 and 4 to agree, and those of pieces 2 and 3 to be the opposite parity: these are the two sides of $7 + 1 = 5 + 3$. This is an elementary six-case calculation, not a packing census. All piece heights in the chosen orientation are even, so the integral-position argument also implies that t is even. □

Proposition 8 (The other hard square)**.** *In an* 840*-square perfect partition with* 20 *pieces, exactly eight of the nine odd-sided shapes are selected. Their orientation and relevant starting coordinate parity belong to the following finite, explicitly specified list. Put*

$$(w_0, \dots, w_8) = (49, 3, 45, 35, 5, 21, 7, 9, 15).$$

Choose one omitted index o*, a set* $X \subseteq \{0, \dots, 8\} \setminus \{o\}$*, and subsets* $P_X \subseteq X$*,* $P_Y \subseteq Y$*, where* $Y = \{0, \dots, 8\} \setminus (\{o\} \cup X)$*, subject to*

$$2 \sum_{i \in P_X} w_i = \sum_{i \in X} w_i, \qquad 2 \sum_{i \in P_Y} w_i = \sum_{i \in Y} w_i. \tag{*}$$

The pieces in X *have odd width and start at* x *of parity* 1 *exactly for indices in* P_X*; the pieces in* Y *have odd height and start at* y *of parity* 1 *exactly for indices in* P_Y*. There are exactly* 44 *such configurations. A quarter-turn allows the additional condition* $|X| \geq 4$*, leaving* 26*. These counts are exhaustible by the displayed nine-bit subset definition; the exact program reconstructs that definition rather than assuming a stored configuration list.*

Proof. There are exactly 22 fitting unordered shapes of area 35280. The nine with an odd side, in the indicated order, are

$$(45, 784), (48, 735), (49, 720), (63, 560), (80, 441),$$
$$(105, 336), (112, 315), (144, 245), (147, 240).$$

Their even partners divided by 16 are precisely the odd numbers w_i. All other shapes have both sides even. Since 20 of 22 shapes are used, at least seven of these nine odd-sided shapes must be used. For either axis, the corresponding alternating-sum equation of Proposition 6, divided by 16 and reduced modulo two, says that the number of selected odd sides on that axis is even. The total selected number is therefore even and hence is exactly eight. The two exact alternating sums are precisely (*): changing all signs has no effect on zero. These necessary equations exhaust the asserted finite list by considering every omitted index and every subset, with no packing inference from a mere compatible configuration.

A quarter-turn exchanges the roles of X and Y (possibly complementing one parity subset, which preserves (*)), so at least four can be horizontal. Reflections in either container midline preserve this orientation condition and preserve the list: a reflection along the odd dimension complements its parity subset and leaves a zero signed sum zero. These two reflections act transitively on the four corners. Thus one may still put a least-index corner shape at the bottom left, using non-strict comparisons if corners coincide. One may *not* also require that corner's width to be at most its height. Exactly this last restriction is omitted in both specialized 840 searches. The same reflection argument applies to the all-odd-width normalization for 21 pieces. This establishes all symmetry reductions used in those programs. □

Whole-program and source boundaries. The independent C++ implementations use a run-length skyline, explicit unordered identities, integer arithmetic, and the stated necessary prunings; the valley and two specialized versions add only the proved valley order and parity restrictions. A deadline produces `UNKNOWN`, never an exclusion. Its source

must be read and a completed exhaustive run must be bound before any finite exclusion is accepted. The independent boundary relaxation can prove infeasibility only; a feasible boundary assignment is not a tiling. The linked repository edition retrieved at commit `5af96dad3646c45e2e0cf9d62481b256c22569a0` is not asserted to be the historical execution revision. In its rectangular routines, a filtered list of orientations need not be closed under reversal: thus dividing its length by two and identifying its reverse index with a rotated piece are not valid in general. This is a current-edition code qualification, not a counterexample to the source's bounded nonexistence assertions. Their complete new finite recovery is supplied in the companion application; scientific adoption remains separate.

References

[1] N. García-Colín, D. Leemans, M. Müßig and É. Roldán, *There is no perfect Mondrian partition for squares of side lengths less than 1001*, arXiv:2311.02385v1, eleven-page fixed source edition (2023).